



Towards Righting CRISPR Code

Everything can be improved – from software code to the code that makes us who we are

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This decade may be more than halfway done, but it will be remembered for one particular decision by the powers that be, and that's to allow scientists to conduct human trials of genetic modification using CRISPR. Finally, some of humanity has overcome its superstitions, stopped pretending that it is perfect, and is looking to dabble with the very code that makes us who we are. Obviously we're talking DNA code, and CRISPR – a new genetic modification tool that's the reason behind the cutesy headline to this article.

CRISPR (Clustered Regularly Interspersed Short Palindromic Repeats) is promising to edit DNA codes with unprecedented precision and simplicity. Why have we made such a big deal about CRISPR in the intro to this article? Because if it succeeds and scientists are able to experiment and analyse the genetic codes of humans, and then change what they want to, it would mean that we as a race would have a near unlimited power of creation. Imagine God (or if you are an atheist, imagine any supreme being) creating life by mixing and inserting lines of code. This code is complex, intractable and was till date not prone to editing, almost like a read only document. Now imagine if we had the key to break

through this code's editing privileges and make changes. CRISPR might just be that key!

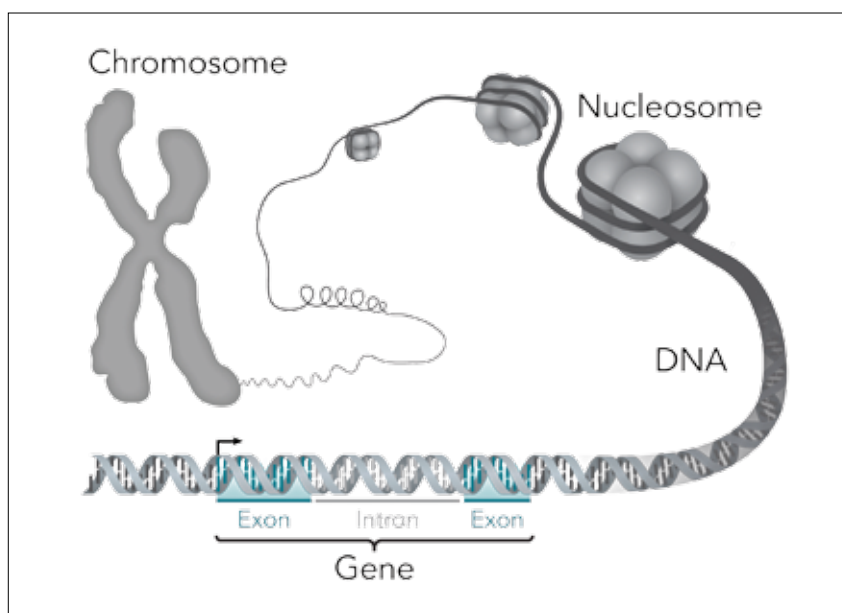
Pre CRISPR days

Genetics is serious business. Even a tiny fault in gene replication can result in serious disorders for an organism, or individual. This happens because a fault in the gene means the wrong proteins are made, or the protein associated with the gene just isn't made at all!

Patients suffering from such deficiencies cannot get the missing protein injected straight into their bloodstream because that comes with

a whole other set of complications. Instead, what gene therapy was doing was delivering a gene into the patient's cells that would then synthesize the missing protein. Using a process called "homologous recombination", the newly introduced DNA which was similar (homologous) to the target DNA, replaced it. Cells which have the correct gene would now be able to produce the missing protein on their own making the cure long lasting, if not completely permanent. So why did we even need an advancement?

We needed one because these therapies were still far from perfect.



Schematic of how genes are bundled up in our chromosomes